CSE7302 - INTERNSHIP

Review- 3 Presentation

Student Details	
Name	BHAGYAVARDHAN
Roll No	20231CSE3038
Section	8CSE09

Under the Supervision of,

Mr.Bikram Sarkar & Ms.Chinmayi G V

Assistant Professor

School of Computer Science and Engineering

Presidency University

Name of the Program: Internship-CSE7302

Name of the HoD: Dr. Asif Mohammed

Name of the Program Internship Coordinator: Dr. Balakrishnan

Name of the School Internship Coordinator: Dr. Md Ziaur Rahman

Content

- About Company or Organization
- Working domain or the technology
- About your team and reporting Manager
- Challenges Faced in Internship
- Objectives of the work
- Literature Review
- Proposed System
- Problem Statement
- System Implementation
- Internship Roadmap

About Company or Organization

- Ankit Automations is a trusted name in the field of security and automation solutions.
- With a focus on innovation, quality, and customer satisfaction, we provide a wide range of products designed to make spaces smarter, safer, and more efficient.
- Join the growing number of clients who trust us to automate their world.

Product Offerings

- Our offerings include boom barriers, sliding gates, automatic doors, garage door, gate motors, flap barriers, swing barriers, turnstiles, CCTV cameras, walkie-talkies, metal detectors, and handheld detectors & etc.

- Whether it's residential, commercial, or industrial, our automation solutions are tailored to meet the unique needs of each client.

Commitment & Support

- Backed by years of experience and a dedicated team, we ensure professional installation, reliable support, and prompt after-sales service.

- At Ankit Automations, we don’t just sell products—we deliver peace of mind and convenience.

Working domain or the technology

- As a **Data Analyst** at Ankit Automations, you'll collect and process data from security systems like CCTV cameras and access control devices to identify patterns in usage and potential vulnerabilities.

- IoT sensors are used to continuously monitor operational parameters like temperature, voltage, uptime, and error frequency.

- Security automation systems such as CCTV cameras, boom barriers, turnstiles, and access control systems form the core application area.

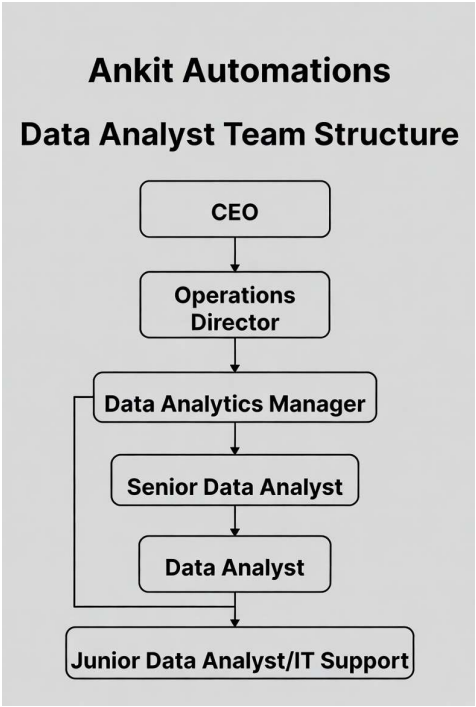
Continue…

- The project operates in the domain of **IoT-enabled smart security automation**, where physical security devices are interconnected through the internet.

- It focuses on **predictive maintenance analytics**, which uses historical and real-time data to forecast failures before they occur.

- Automation analytics enables proactive decision-making rather than reactive maintenance.

About your team and reporting Manager



Continue…

- The internship was conducted at **Ankit Automations**, a company specializing in security and automation solutions.
- The team includes automation engineers responsible for hardware deployment.
- The reporting structure helps interns understand real-world organizational workflows.
- The team follows professional communication and reporting standards.

Challenges Faced in Internship

- Adjusting to cross-functional team dynamics.
- Adapting to ambiguous, unstructured real-world data.
- Communicating insights to non-technical teams.
- Dealing with time-zone or hybrid work confusion.
- Data preprocessing consumed significant time and effort.

Objectives of the work

- To analyze device behavior using analytics techniques.
- To understand maintenance challenges faced in security infrastructures.
- To apply predictive maintenance concepts in a real-world scenario.
- To collect real-time operational and environmental data.

Continue….

- Transform raw device data into strategic assets overall, turning logs from IoT-connected products into meaningful intelligence that aligns with the company's mission of innovation, quality, and customer-focused automation.

- Support business expansion by analyzing market and usage data to identify high-demand sectors (e.g., industrial vs. commercial) and inform strategic decisions.

Literature Review

- Research studies highlight predictive maintenance as superior to reactive methods.
- Literature shows IoT as a key enabler for smart monitoring systems.
- Existing systems lack integration between IoT and analytics.
- Research identifies data quality as a major challenge.

Proposed System

- The proposed system integrates IoT sensors with security devices.
- Data is transmitted securely to cloud servers.
- Machine learning models analyze trends and anomalies.
- Maintenance schedules are optimized using analytics.
- The solution improves system reliability and uptime.

Problem Statement

- Existing maintenance methods are reactive.
- Manual monitoring is inefficient and error-prone.
- Maintenance costs increase due to emergency repairs.
- Security automation systems face unexpected operational failures.
- Maintenance schedules are not data-driven.

Hardware and Software Details

- **Laptops/Workstations:** Development team using Windows 10/11 laptops with 16GB RAM, Intel i7 processors, and dual monitors for productivity.

- **Development Tools:** Visual Studio Code for coding, GitHub for version control and collaboration, Docker for containerization ensuring consistent environments, and Postman for API testing.

- **ETL Tools:** Apache Airflow for workflow orchestration and scheduling, custom Python scripts using SQLAlchemy for database connectivity, and Pandas for data transformation logic.

Continue….

- **Communication Tools:** Slack for team collaboration, Microsoft Teams for meetings, and JIRA for project task tracking and sprint planning.

- **Monitoring Tools:** AWS CloudWatch for infrastructure monitoring, Grafana for custom metrics dashboards, and PagerDuty for incident alerting and escalation management.

- **Backup Systems:** AWS S3 with versioning enabled, automated backup scripts, and tested disaster recovery procedures with quarterly restoration drills.

Advantages of Proposed System/Work

- Enables early fault detection.
- Reduces manual intervention.
- Extends equipment lifespan.
- Optimizes maintenance scheduling.
- Enhances customer satisfaction.

Internship Road Map

Note: Write in the below table what u will be achieving in each review

Review 1	Review 2	Review 3	Review 4	Review 5
Learning domain basics for security & automation: Understand boom barriers, turnstiles, flap barriers, CCTV metadata, access patterns (domain knowledge, physical security systems, access control devices)	Building an foundational knowledge in time-series & IoT data concepts: sensor logs, anomaly detection, predictive maintenance basics (time-series analysis, IoT sensor data detection.	The system enhances decision-making by providing data-driven insights, helping organizations plan maintenance activities more effectively and efficiently.		

Github Link

- https://github.com/bhagyavardhanIntern08

Q&A

Thank you !!